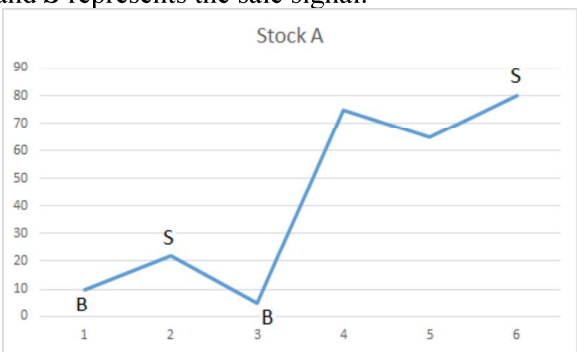
Problem 03: Stock Trader's Dilemma

Time limit: 1 second

These days' stock markets are doing amazing. Investment is very easy, and anyone can become an active trader. Upon looking at such news, CM (Capital Makers) decided to invest in the stock exchange. Having no prior experience, they are in deep trouble in timing the buy and sell signals. They are losing the money of their investors and need your help to come up with a plan to ace the stock market and make profits by buying and selling a stock.

The strategy is to buy a stock, hold it till it gives profit and then sell it. Afterwards, buy the stock once again and sell it. However, since they had already made a lot of mistakes, their shareholders do not trust them anymore. They are restricting them to deal only in one stock at a time and limit the transaction to at most '***k***'. Such that the stock can only be bought '***k***' times and sold '***k***' times. By allowing '***k***' transaction, the shareholders want to have maximum profit.

For example, given the trading data of a stock '***A***' as ***[10, 22, 5, 75, 65, 80]*** and ***k = 2***, the maximum profit can be earned when the stock is bought on day ***1***, sold on day ***2***. Then bought on day ***3*** and sold on day ***6***. Thus, the complete transaction looks like, ***(22-10) + (80-5) = 87***. In the image below, ***B*** represents the buy signal, and ***S*** represents the sale signal.



Input

The first line of the input consists of ***t, (1 ≤ t ≤ 50)*** representing the total number of test cases given. Each test case is represented in three lines. The first line of the test cases represents, ***'k' (0 ≤ k ≤ 10^3)*** allowed trades for the session. The next line contains ***'n' (1 ≤ n ≤ 10^4)*** number of days. The next line contains ***'n'*** values, space separated, representing stock prices on ***'n'*** days.

Output

Output consists of t lines, each line contains exactly one number, representing the maximum profit of the trade.

Sample Input	**Sample Output**
3 2 6 10 22 5 75 65 80 3 6 10 22 5 75 65 80 1 5 90 80 70 60 50	87 97 0